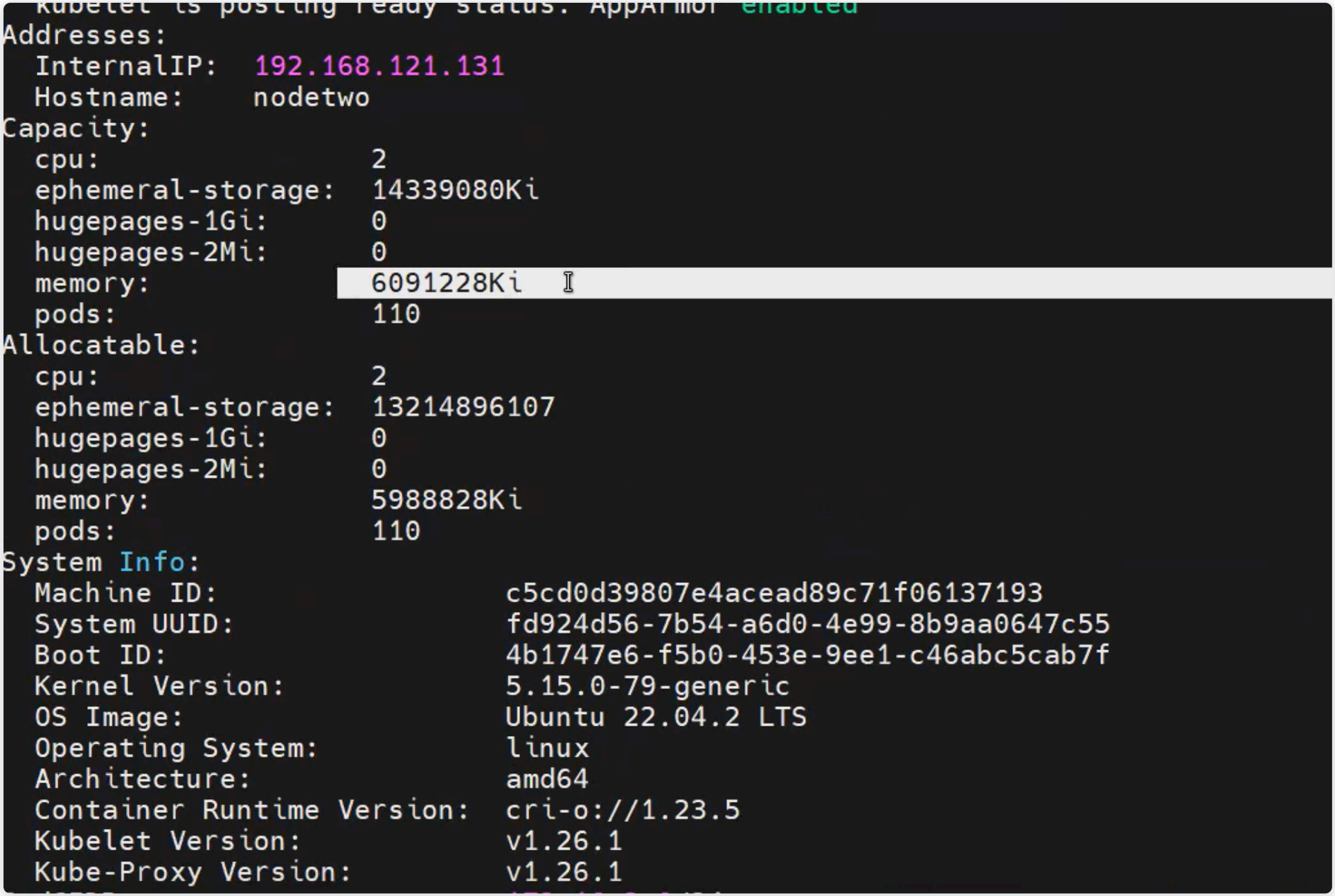
KUBERNETES CKA PREP › MY LEARNING NOTES › SESSION 02

Screenshots: Quality of Service (QoS)

Raw screenshots captured while watching this session, kept alongside the written notes (02-quality-of-service.md) for visual reference during interview revision.

10 screenshots · captured 2026-09-10, 3:24 PM - 4:37 PM

```
root@manager:~/weekends# cat yourpod.yaml
---
apiVersion: v1
kind: Pod
metadata:
  name: frontend
  labels:
    abc.com/name: nginx
    abc.com/managed-by: manual
    abc.com/version: "1"
    abc.com/component: frontend
    abc.com/part-of: webserver
    abc.com/instance: august
spec:
  containers:
  - name: nginx
    image: nginx
    ports:
    - containerPort: 80
```

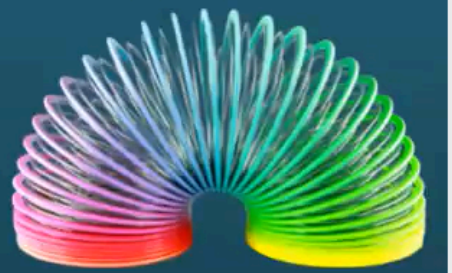

Kubelet is posting ready status. AppArmor enabled
Addresses:
InternalIP: 192.168.121.131
Hostname: nodetwo
Capacity:
cpu: 2
ephemeral-storage: 14339080Ki
hugepages-1Gi: 0
hugepages-2Mi: 0
memory: 6091228Ki
pods: 110
Allocatable:
cpu: 2
ephemeral-storage: 13214896107
hugepages-1Gi: 0
hugepages-2Mi: 0
memory: 5988828Ki
pods: 110
System Info:
Machine ID: c5cd0d39807e4acead89c71f06137193
System UUID: fd924d56-7b54-a6d0-4e99-8b9aa0647c55
Boot ID: 4b1747e6-f5b0-453e-9ee1-c46abc5cab7f
Kernel Version: 5.15.0-79-generic
OS Image: Ubuntu 22.04.2 LTS
Operating System: linux
Architecture: amd64
Container Runtime Version: cri-o://1.23.5
Kubelet Version: v1.26.1
Kube-Proxy Version: v1.26.1

KUBERNETES

HAS TWO TYPES OF RESOURCES

COMPRESSIBLE

LIKE CPU. KUBERNETES CAN GIVE AND TAKE CPUS WHENEVER IT LIKES. POD THAT NEED CPU AND DON'T GET IT WILL WAIT.

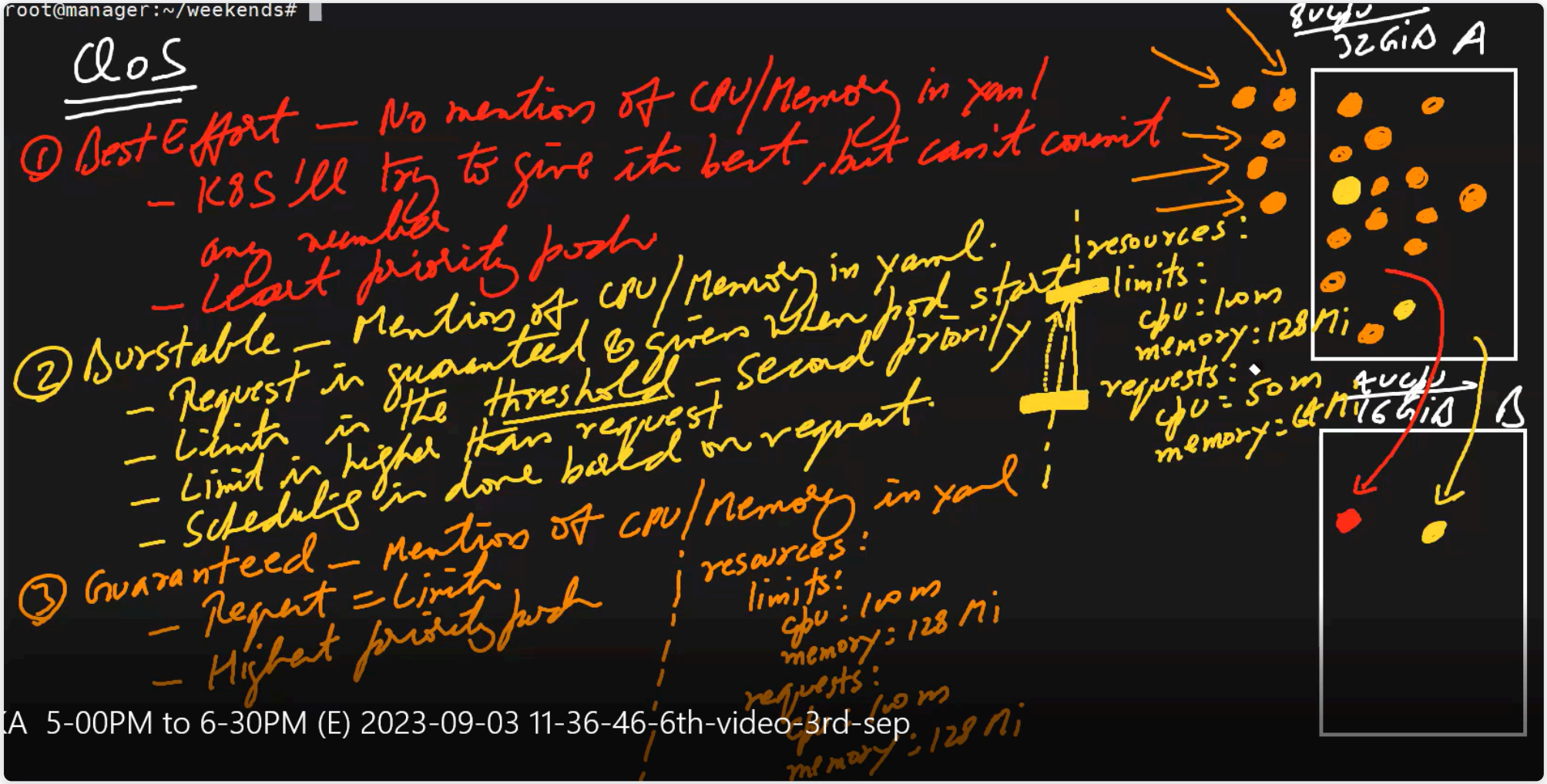


NON COMPRESSIBLE

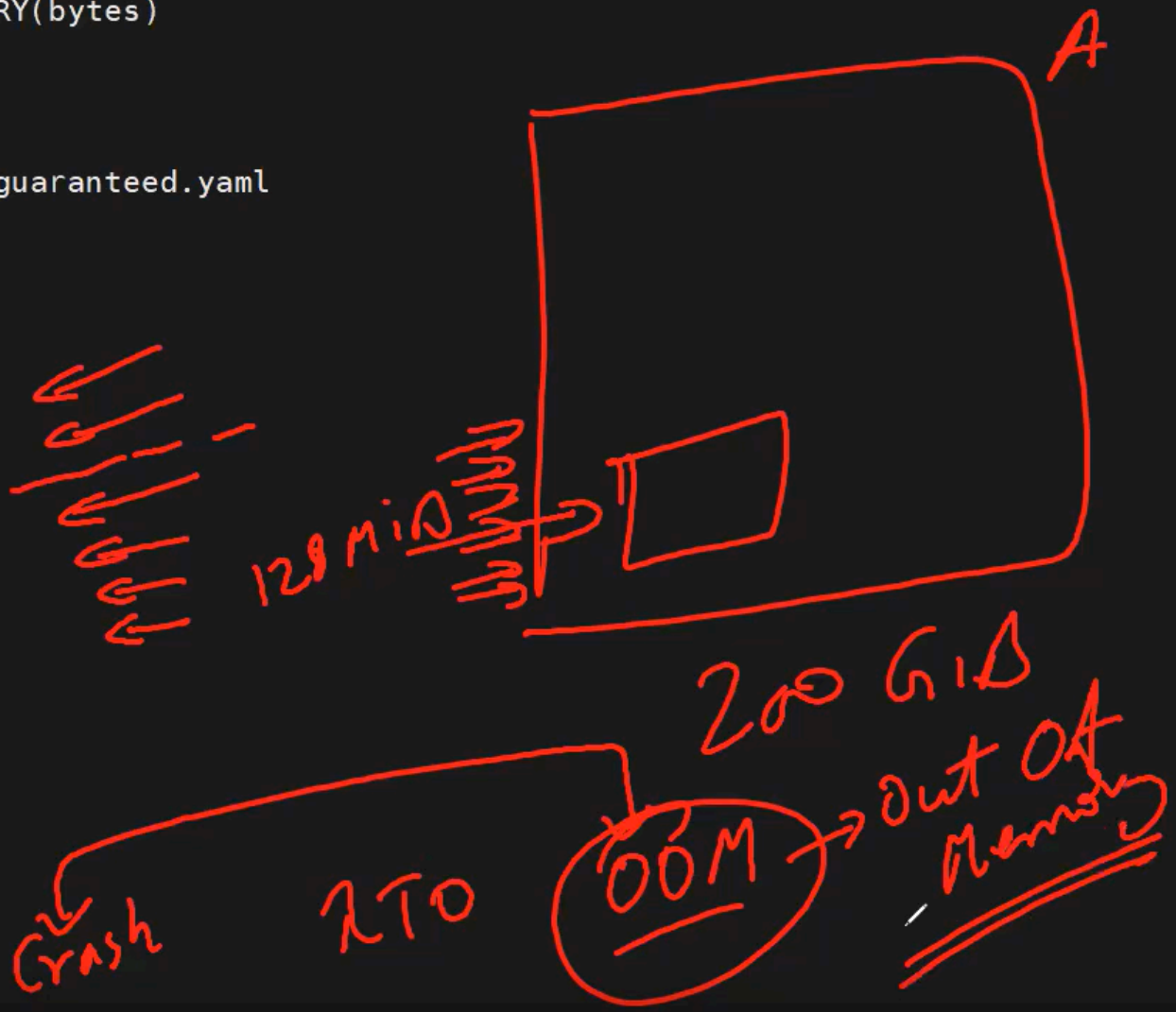
LIKE MEMORY. KUBERNETES CAN'T TAKE IT AWAY WITHOUT KILLING THE POD.



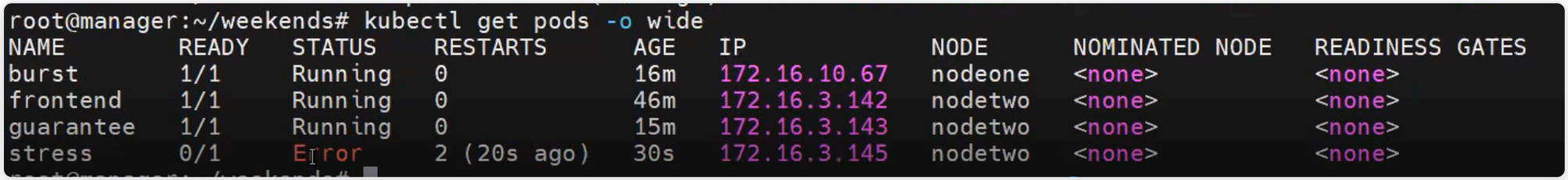
```
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
Volumes:
  kube-api-access-zfkmg:
    Type:                     Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:   3607
    ConfigMapName:            kube-root-ca.crt
    ConfigMapOptional:        <nil>
    DownwardAPI:              true
QoS Class:                    BestEffort
Node-Selectors:               <none>
Tolerations:                  node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                              node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type     Reason      Age    From                Message
```


root@manager:~/weekends#
QoS
① Best Effort – No mentions of CPU/Memory in yaml
– K8S will try to give its best, but can't commit
– Least priority pods
② Burstable – Mentions of CPU/Memory in yaml.
– Request is guaranteed & given when pod start
– Limits in the threshold – second priority
– Limit is higher than request
– Scheduling is done based on request.
③ Guaranteed – Mentions of CPU/Memory in yaml
– Request = Limits
– Highest priority pods
resources:
limits:
cpu: 100m
memory: 128Mi
requests:
cpu: 50m
memory: 64Mi
resources:
limits:
cpu: 100m
memory: 128Mi
requests:
cpu: 100m
memory: 128Mi
8vCPU
32GiB A
8vCPU
16GiB B
A 5-00PM to 6-30PM (E) 2023-09-03 11-36-46-6th-video-3rd-sep

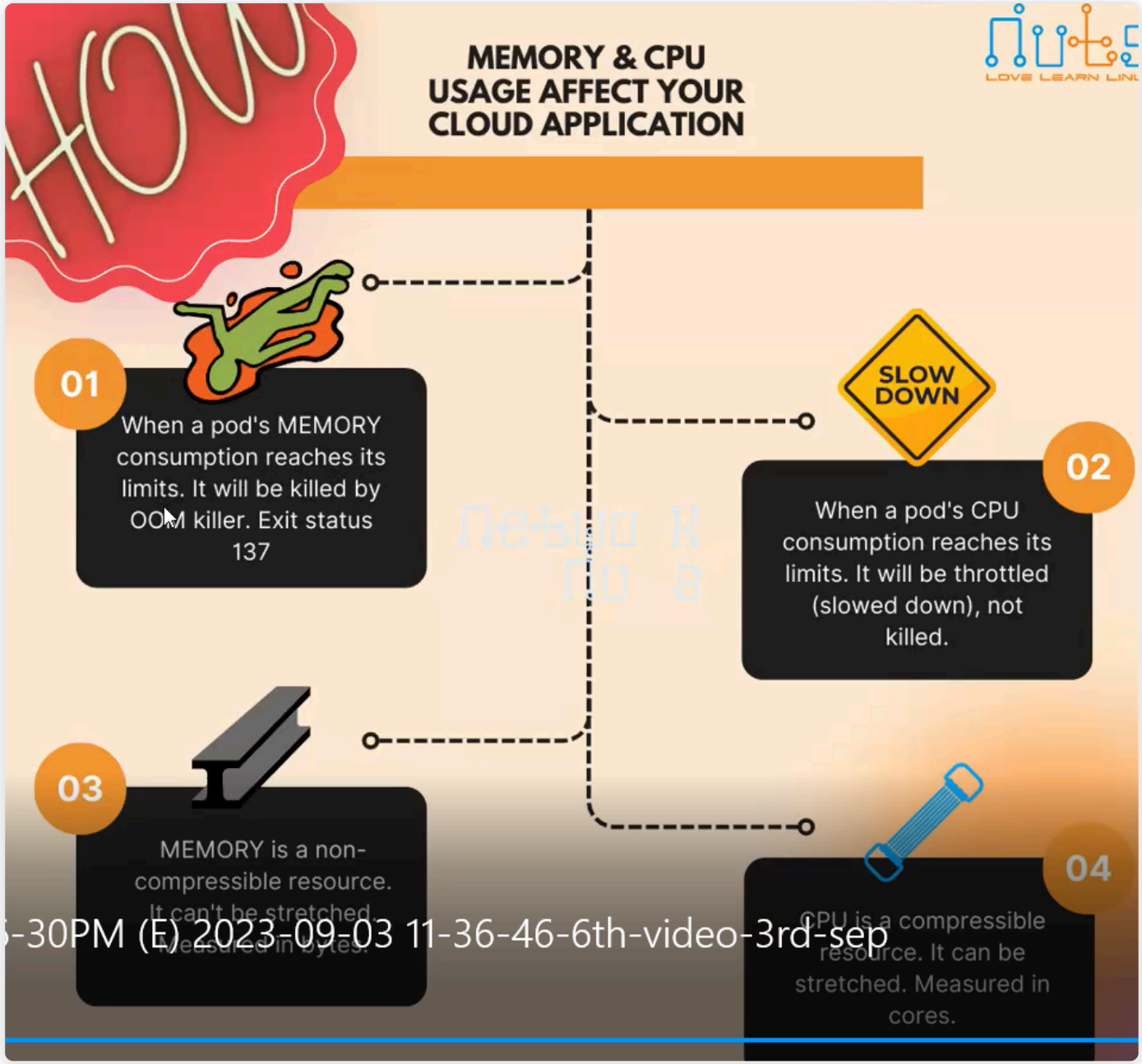
```
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends# kubectl top pods
NAME         CPU(cores)   MEMORY(bytes)
burst        0m           6Mi
frontend     0m           8Mi
guarantee    0m           2Mi
root@manager:~/weekends#
root@manager:~/weekends# cat guaranteed.yaml
apiVersion: v1
kind: Pod
metadata:
  name: guarantee
spec:
  containers:
  - name: qos-demo-ctr
    image: nginx
    resources:
      limits:
        memory: "128Mi"
        cpu: "100m"
      requests:
        memory: "128Mi"
        cpu: "100m"
root@manager:~/weekends#
```



```yaml
apiVersion: v1
kind: Pod
metadata:
  name: stress
  namespace: learning
  labels:
   app: nnappone
spec:
  containers:
    - name: networknuts-app
      image: lovelearnlinux/stress:latest
      resources:
        limits:
          memory: "250Mi"
        requests:
          memory: "150Mi"
      command: ["stress"]
      args: ["--vm", "1", "--vm-bytes", "275M", "--vm-hang", "1"]
~
~
~
```


root@manager:~/weekends# kubectl get pods -o wide
NAME READY STATUS RESTARTS AGE IP NODE NOMINATED NODE READINESS GATES
burst 1/1 Running 0 16m 172.16.10.67 nodeone <none> <none>
frontend 1/1 Running 0 46m 172.16.3.142 nodetwo <none> <none>
guarantee 1/1 Running 0 15m 172.16.3.143 nodetwo <none> <none>
stress 0/1 Error 2 (20s ago) 30s 172.16.3.145 nodetwo <none> <none>


Memory
Container
kill
restart
CPU
CPU Throttling
slows down
to 6-30PM (E) 2023-09-03 11-36-46-6th-video-3rd-sep


HOW
MEMORY & CPU USAGE AFFECT YOUR CLOUD APPLICATION
01
When a pod's MEMORY consumption reaches its limits. It will be killed by OOM killer. Exit status 137
SLOW DOWN
02
When a pod's CPU consumption reaches its limits. It will be throttled (slowed down), not killed.
03
MEMORY is a non-compressible resource. It can't be stretched. Measured in bytes.
04
CPU is a compressible resource. It can be stretched. Measured in cores.
-30PM (E) 2023-09-03 11-36-46-6th-video-3rd-sep